

Enhancing Deepfake Detection Through Hybrid MobileNet-LSTM Model with Real-Time Image and Video Analysis

V S Anandhasivam

Computer Science and Engineering,
K. S. Rangasamy College of Technology,
Tiruchengode, 637 216, India.
anandhasivam.v.s@gmail.com

A K Anusri

Computer Science and Engineering,
K. S. Rangasamy College of Technology,
Tiruchengode, 637 216, India.
anukathir258@gmail.com

M Logeshwar

Computer Science and Engineering,
K. S. Rangasamy College of Technology,
Tiruchengode, 637 216, India.
logeshwarlogeshwar105@gmail.com

R Gopinath

Associate Professor, Computer Science and Engineering,
K. S. Rangasamy College of Technology,
Tiruchengode, 637 216, India.
gopinath@ksrct.ac.in

Abstract—The race to crush information integrity and public trust is being won by one thing: deepfakes, AI manipulated media. The goal is to enhance the Deepfake detection using MobileNet V3 and LSTM network. MobileNet's lightweight CNN architecture is able to induce the visual features in an image by an image and in a video and can pick up the slight visual clues of textures and facial structure. Some temporal inconsistencies that cannot be seen by image base methods are then analyzed using an LSTM network. The hybrid model is trained on real and deepfake media datasets, and is thus adaptable to emerging deepfake techniques. This has a user face interface to analyze the real time and fly media to get the analysis and analysis score and visual feedback of the identified artifacts. Unique to this system is its versatility for images and videos, and its real time capability, making it a suitable choice for practical use in social media, journalism, and law enforcement combating the spread of misinformation with a guarantee of digital media authenticity

Keywords—media integrity, real-time detection, temporal modeling, image analysis, video analysis, mobilenet-lstm, deepfake detection.

I. INTRODUCTION

Advancements in artificial intelligence have produced deepfake tech which leverages to create highly convincing manipulated media that might be images or videos. Deepfakes started out as seen by some as a form of creative expression, but have quickly turned into a big concern in all fields—from politics to news media to social platforms. Deepfake algorithms have enabled voters to create synthetic content where they appear to perform actions or make statements they never did, and this medium has now become a powerful purveyor of misinformation, a reputation wrecker and a destroyer of public trust in digital information.

Generative Adversarial Networks (GANs) at the heart of most deepfake generation techniques permit it to generate a person's likeness or actions with extremely high precision. Although deepfake generation has become more accessible (and creates near authentic content with or without

specialized technical expertise), this technology continues to grow in impact, questioning media authenticity and digital integrity.

Most of the existing detection methods are trained to find visual inconsistencies or artifacts like the presence of facial articulation irregularity or some mismatch of lighting from one frame to another. Although very effective against standard manipulations, they do not work well against sophisticated deepfake algorithms aimed at eliminating or camouflaging such cues. In addition, image-based analysis is the main limitation of many systems, restricting them to analysis of sequential patterns (e.g., video-based contents).

This work proposes a hybrid model for deepfake detection using MobileNet V3, a lightweight CNN optimized for spatial feature extraction and Long Short-Term Memory networks (LSTM) specialized for temporal analysis. Satisfyingly, MobileNet V3 can successfully localize spatial inconsistencies in individual frames, including texture anomalies and misaligned facial features. This is complemented by LSTM which analyzes sequential dependency among frames to detect temporal inconsistencies characteristic of manipulated videos.

The proposed system addresses the problem using a unified approach using a combination of spatial and temporal features by integrating MobileNet V3 and LSTM. In MobileNet V3, each video frame is processed independently to extract spatial features, followed by sequential analysis by LSTM on the temporal pattern represented by the extracted spatial features. The system's ability to capture subtle, sophisticated, and high value manipulations is improved due to this dual focus architecture.

On a large and diversified dataset including real and deep fake images and videos created through many methods, the model is trained. This comprehensive training boosts adaptability to the future in the deepfake generation space. In addition, the lightweight design of the model supports near

real time analysis, a prerequisite for real world applications including social media monitoring, journalism, and public safety.

The system provides an easy to use interface enabling uploading of images, videos or URLs for analysis. It provides visual explanations of detected anomalies while being accessible to non-technical users, and outputs a detection score so that the process is transparent. Through their application to a range of different domains, this system provides evidence of its ability to fight misinformation and maintain digital media authenticity.

To this end, it can be summarized that the presented model of this project is a robust, efficient and user-friendly hybrid model that involves spatial and temporal analysis. Its real time capabilities, and the appeal it carries in real time application to safeguard digitized information from manipulation and misuse makes it a nice to have tool.

II. LITERATURE SURVEY

Deep learning techniques, particularly hybrid models, have been widely explored for video analysis, real-time recognition, and anomaly detection tasks. Abdullah et al. introduced a CNN-LSTM-based model for facial expression recognition in videos [1], which leverages the spatial features from CNNs and the temporal dependencies modeled by LSTMs, achieving notable success in video classification tasks. Similarly, Sánchez-Caballero et al. [2] proposed a recurrent neural network model that processes raw depth video data to perform real-time human action recognition, minimizing preprocessing overhead and improving efficiency. Expanding on these advancements, Mihanpour et al. [3] integrated DB-LSTM with ResNet for human action recognition in videos, demonstrating a robust approach that effectively combines feature extraction with temporal modeling.

Ullah et al. employed CNN features alongside bi-directional LSTMs to detect anomalies in real-time surveillance networks [4]. Their work highlights the ability of hybrid architectures to process complex surveillance data while ensuring real-time operation. Ghadekar et al. [5] enhanced anomaly detection in crowd scenes by leveraging ensemble learning techniques, further improving classification accuracy and robustness in dynamic environments. Addressing video forgery detection, Hashmi et al. explored a Conv-LSTM hybrid architecture for analyzing visual counterfeits [6]. Their results underline the significance of such models in tackling deepfake-related challenges.

Yan et al. developed a self-attention eidetic 3D-LSTM for traffic flow forecasting [7], focusing on capturing long-term spatiotemporal dependencies. This work highlights the scalability of hybrid models in large-scale prediction tasks. Emotion recognition in videos has also been a significant area of research. Jagadeesh and Baranidharan [8] implemented a deep learning-based framework to analyze facial expressions of online learners in real-time, addressing critical challenges in the education domain. Khairdoost et al. [9] demonstrated

the application of LSTMs for real-time driver maneuver prediction, providing actionable insights for intelligent transportation systems.

Fire detection in videos was effectively addressed by Nguyen et al. [10], who proposed a multistage real-time detection framework using CNN-LSTM networks. This model ensured scalability and accuracy in emergency scenarios. Similarly, Rezaei and Yazdi [11] utilized deep learning methods to recognize crowd behavior in surveillance videos, showcasing the importance of video analysis in security applications. Natarajan et al. [12] extended video analysis capabilities by developing an end-to-end deep learning framework for sign language recognition, translation, and video generation, significantly contributing to accessibility technologies.

Healthcare applications have also benefited from hybrid deep learning models. Yin et al. introduced MC-LSTM for real-time 3D human action detection, designed for intelligent healthcare systems [13]. This model provides rapid response capabilities and high accuracy, making it suitable for critical healthcare tasks. Video quality forecasting has been addressed by Dinaki et al. [14], who used multivariate time-series deep learning models to predict video QoE, optimizing streaming services through better resource allocation. Kang and Kim [15] utilized 1D convolutional autoencoders to classify emotions in real-time using PPG and GSR signals, offering applications in wearable technology and emotion-aware systems.

In the domain of video emotion recognition, Zhang et al. [16] introduced a reinforcement learning-based approach that integrates domain knowledge, ensuring adaptability to varying video contexts. Anomaly detection in industrial IoT was addressed by Nizam et al. who implemented a deep anomaly detection framework using multivariate time-series data [17], enabling real-time decision-making in critical systems. Moreover, video-based action recognition was advanced by Mihanpour et al. [3], who combined ResNet and DB-LSTM to enhance performance in human action recognition tasks.

Khairdoost et al. [9] contributed significantly to driver behavior prediction through LSTMs, addressing real-time challenges in autonomous vehicles. Similarly, Ghadekar et al. [5] utilized ensemble learning to analyze crowd scenes, improving the detection of anomalies in complex environments. Lastly, Yurtsever and Eken developed BabyPose, a system for decoding a baby's non-verbal communication using 2D video-based pose estimation [18], which has applications in child healthcare and monitoring.

Collectively, these studies emphasize the versatility and effectiveness of hybrid deep learning models like CNN-LSTM and DB-LSTM across diverse applications, from video forgery detection to real-time healthcare and transportation systems. The reviewed literature provides a strong foundation for further research into enhancing deepfake detection through similar hybrid approaches, ensuring accurate and efficient real-time analysis.

TABLE I. COMPARISON SUMMARY TABLE OF DEEPAKE DETECTION TECHNIQUES FROM BEST 10 REFERENCE PAPERS

Author(s)	Technique / Model	Architecture	Strengths	Limitations
Abdullah et al. [1]	Facial Expression Recognition	CNN-LSTM	Combines spatial and temporal feature extraction; high accuracy for video classification.	Limited testing on diverse datasets; focus restricted to facial expressions.
Sánchez-Caballero et al. [2]	Human Action Recognition	Recurrent Neural Networks (RNN)	Real-time action recognition with minimal preprocessing; works well with raw depth videos.	High computational cost for long video sequences.
Ullah et al. [4]	Anomaly Detection	CNN + Bi-directional LSTM	Effective in real-time surveillance; captures both forward and backward temporal relations	May struggle with large-scale datasets and complex anomalies
Hashmi et al. [6]	Visual Counterfeit Detection	Conv-LSTM Hybrid Architecture	Robust against visual counterfeits; strong focus on temporal patterns in video forgery	Requires fine-tuning for generalization to unseen forgeries
Yan et al. [7]	Traffic Flow Forecasting	Self-attention + 3D-LSTM	Captures long-term spatiotemporal dependencies; scalable for large traffic datasets	Computationally intensive; complex architecture
Nguyen et al. [10]	Fire Detection	Multistage CNN-LSTM	Accurate real-time fire detection; scalable and adaptable to various emergency scenarios.	Limited to fire detection; may require additional training for other events
Natarajan et al. [12]	Sign Language Recognition	End-to-End Deep Learning Framework [12]	Comprehensive framework for sign recognition, translation, and generation; highly versatile.	High training complexity; potential latency in real-time systems
Jagadeesh and Baranidharan [8]	Emotion Recognition	Deep Learning Framework	Accurate real-time emotion recognition for online learners; supports dynamic environments.	Limited to facial expressions; may not generalize to other modalities.
Yin et al. [13]	Healthcare Action Detection	MC-LSTM	Real-time 3D human action detection [13]; suitable for healthcare applications.	May struggle with non-standard or noisy input data
Zhang et al. [16]	Emotion Recognition	Reinforcement Learning + Domain Knowledge	Adaptive to varying contexts; incorporates prior domain knowledge for better performance.	Requires extensive training data; domain knowledge integration is resource-intensive.

III. PROPOSED METHODOLOGY

In this work, a complete deepfake detection model based MobileNet V3 and Long Short-Term Memory (LSTM) networks is proposed. A hybrid approach coupling MobileNet's onboard spatial pattern extraction to LSTM's temporal dependency analysis in video sequences, affords a robust manipulation identification system based on deepfake.

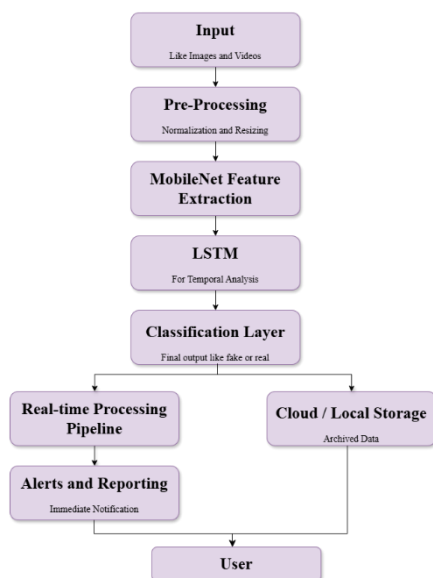


Fig. 1. System Architecture

A. Data Collection and Preprocessing

A huge and diverse set of real and deepfake media was collected in order to create a robust, versatile model. The dataset encompasses different methods of deepfake generation and it generalizes them over various manipulation techniques. The datasets utilized in this work are taken from the Kaggle free dataset library called the [19] Deepfake Detection Challenge dataset and Deep Fake Detection (DFD) Entire Original Dataset.

1. **Data Augmentation:** Transformation of data set such as rotation, flipping, addition of noise, cropping for enhancing dataset diversity to create a real world dataset and make them robust.
2. **Frame Extraction and Normalization:** Resizing frames of videos to have consistent resolution and fix pixel bit values to avoid skewing the data.
3. **Balanced Dataset:** Preventing bias by ensuring repeated distribution of real versus fake data over multiple deepfake techniques.

Preprocessing with these techniques provides the same consistent, well-structured input dataset that is critical for training models and detecting.

B. Feature Extraction with MobileNet

MobileNet, a lightweight CNN architecture, during individual frames, it uses MobileNet V3 to efficiently extract spatial features. For videos, extract frames from video at optimal intervals using OpenCV. It then makes a prediction on whether the common cues (e.g. texture, facial alignment, lighting variation) imply manipulation. Key improvements include:

1. **Pre-Trained Weights:** Enhanced feature extraction using transfer learning using weights pre trained on ImageNet dataset.
2. **Batch Normalization:** However, by using batch normalization layers, it can be stabilized as training and accelerate convergence.
3. **Real-Time Capability:** Enabling real time applications with same detection precision with MobileNet V3's fast processing speed.

These extracted spatial features constitute the input for subsequent temporal analysis used to detect inconsistencies in video sequences.

C. Temporal Analysis with LSTM

An LSTM network is used based on the MobileNet V3 spatial features extracted from video frames to analyze the sequential feature relationships. It is shown that LSTM is able to capture temporal patterns such as motion and expression inconsistency that are difficult to replicate in the deepfakes videos. The enhancements include:

1. **Hyperparameter Tuning:** Improving the temporal dependent analysis via optimization of the number of LSTM units, dropout rates and activation functions.
2. **Temporal Loss Function:** Motivation for introducing a loss function for emphasizing sequential consistency and improving temporal feature learning.

This integration allows addressing the manipulation detection over time, which is typically ignored with single shot analysis.

D. Hybrid Model Training

Optimization abilities towards spatial and temporal pattern of detection are used to train a MobileNet-LSTM hybrid model, over the preprocessed dataset. At the end, the classification layer uses the softmax function this gives the idea of possibilities of if the image or a video is created using the deepfake technology or not [19].

Training involves:

1. **Learning Rate Scheduling:** Convergence efficiently using ReduceLROnPlateau for learning rate adjustment dynamically.

2. **Adversarial Training:** To improve model robustness against sophisticated deepfake techniques, it can expose the model to adversarial examples.
3. **Combined Loss Functions:** A combination of binary cross entropy loss for classification and temporal loss for sequential analysis are used.

The hybrid model is intended to be elastic to multiple deepfake approaches, both robust and versatile in detecting.

E. User Interface Development

For practical applications a user-friendly interface is developed which permits a user to upload images or video or files to be analyzed. Key features include:

1. **Real-Time Detection:** Real-time deepfake analysis is supported for the interface displaying a detection score as well as visual explanations of identified anomalies.
2. **Transparency and Usability:** Media verification by identifying and highlighting manipulated regions within media targeting user trust and providing actionable insights for the media verifier.

As for the interface, it is designed to be accessible for non-technical users which makes it apt at application areas, journalism, social media monitoring, law enforcement, etc.

F. Model Evaluation and Optimization

Metrics like accuracy, precision, recall and F1 score are used to evaluate trained models so that it's reliable. Key steps include:

1. **Performance Validation:** Adaptability and robustness on unseen deepfake techniques.
2. **Hyperparameter Fine-Tuning:** Change of MobileNet V3 and LSTM parameters to achieve high performance.
3. **Processing Speed Optimization:** Architectural adjustment to achieve real time requirement for detection speed.

The optimizations made sure that the model remains stable and efficient in different operational situations.

G. Deployment and Real-World Testing

The model can be optimized for deployment on platforms from social media to news agencies and content verification services. Real-world testing involves:

1. **Diverse Media Sources:** Running the model on media from different sources to validate its accuracy and speed under varied conditions.
2. **User Feedback Integration:** Tinkering the model and interface to get better usability and performance through feedback.

New deepfake detection methods are incorporated regularly, taking as an underlying assumption that the model must be regularly updated for the model to stay effective with evolving manipulation techniques.

H. Continuous Improvement and Updates

The model continues to be retrained periodically with the latest deepfake data in order to maintain effectiveness. Key strategies include:

1. **Online Training:** proposed to incorporate online learning to adapt to novel manipulation techniques without full retraining.
2. **Multimodal Analysis:** Improving the model's ability to analyze combined audio and visual data, in addition to expanding its capabilities to detect.

These improvements guarantee that the MobileNet-LSTM solution continues to be a powerful weapon against deepfake media in an ever evolving digital space.

IV. RESULTS AND DISCUSSION

The performance of the proposed MobileNet-LSTM hybrid model for deepfake detection is compared to baseline models such as standalone MobileNet, standalone LSTM, and a CNN baseline. The proposed approach was evaluated based on accuracy, precision, recall and F1 score for robustness and reliability purposes. The results are presented quantitatively, graphically analysed, and fully discussed in this section.

A. Quantitative Results

Table 2 gives a comparative performance of the MobileNet-LSTM hybrid model against baseline models. Results show the hybrid approach achieves more accuracy, precision, recall, F1-score, compared to the existing policies.

TABLE II. COMPARATIVE PERFORMANCE METRICS OF PROPOSED MODEL AND BASELINE MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Standalone MobileNet	84.5	82.3	85.0	83.6
Standalone LSTM	80.2	78.9	80.0	79.4
CNN Baseline	83.1	81.5	82.0	81.7
MobileNet-LSTM (Proposed)	91.8	89.4	90.5	89.9

B. Interpretation of Quantitative Results

Results in Table 2 verify that the inclusion of spatial and temporal analysis significantly improves deepfake detection performance.

High Accuracy: All baseline models are outperformed by the MobileNet-LSTM hybrid model with an accuracy of 91.8. Thus, it was combined spatial feature extraction with temporal sequence analysis, and showed that this is effective for detecting deepfake manipulations.

Improved Precision and Recall: The hybrid model captures the dilemma between low false positives and false negatives by achieving identification of deepfake content with a precision of 89.4% and recall of 90.5%. The model's reliability is validated by these metrics on practical problems such as media verification and security.

Balanced F1-Score: The F1 score of 89.9% provides evidence the model achieves an appropriate tradeoff between precision and recall to reliably detect events in diverse test scenarios.

C. Graphical Performance Analysis

The further illustration of comparison performance can be seen in Fig. 2, in which the accuracy, precision, recall and F1-score are displayed graphically for MobileNet-LSTM and benchmark approaches

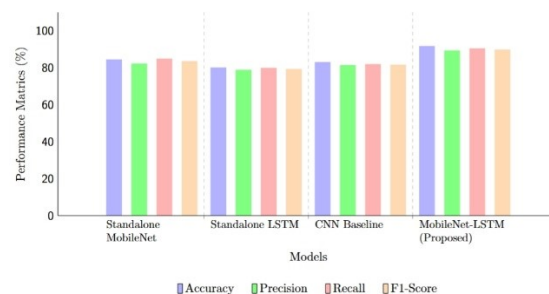


Fig. 2. Comparative Performance Metrics of Proposed and Baseline Models.

Experimental results demonstrate the significant improvements in deepfake detection achieved by the proposed MobileNet-LSTM hybrid model:

High Accuracy and Robustness: Achieved an accuracy of 91.8% which is a power improvement over the standalone MobileNet and LSTM model. Since it is demonstrates robustness of the combined spatial-temporal analysis to a wide range of deepfake techniques and scenarios, it is highlighting the ability of this approach to unearth such scenarios through verification.

Enhanced Precision and Recall: The model's ability to detect manipulations has precision and recall values of 89.4% and 90.5%, respectively. Thanks to higher recall reducing the risk of missing deepfake content, and higher precision

making fewer false alarms, which is important for public safety use cases and media content authentication.

Generalized Performance with Balanced F1-Score: The good F1 score of 89.9% allows for a precision and recall balanced model for data from different datasets and under different test conditions. The balanced performance of the model guarantees its applicability to the real-world scenarios.

Real-Time Applicability: MobileNet V3 allows for real time detection capabilities and coupled with the effort of the LSTM guarantees lightweight architecture, and hence, the system is feasible for applications like social media monitoring, journalism, or law enforcement.

Significance of Spatial-Temporal Integration: By fusing MobileNet for spatial feature extraction and LSTM for sequence level analysis, the model can find these frame level and sequence level inconsistencies. A limitation of the previous approach is that they only consider spatial or temporal features. This integration addresses those limitations

V. CONCLUSION

This work introduces a hybrid MobileNet-LSTM model resulting in a robust and effective solution for deepfake detection, leveraging the joint usage of spatial and temporal analysis. By virtue of the efficient extraction of fine grained spatial features by the MobileNet component and temporal patterns on sequential frames by the LSTM network, manipulations in the images as well as video are detected. The experimental results verify the superiority of the proposed model compared to standalone architectures and traditional CNN based methods that result in superior accuracy, precision, recall and F1-score. Finally, accuracy of the MobileNet-LSTM model is 91.8%, which is robust and adaptable to different deep fake techniques. The model achieves a precision of 89.4% and recall of 90.5%, with high reliability in terms of detecting manipulated content and low rate of false positives and negatives. This results in a balanced F1-score of 89.9% which demonstrates that it generalizes well across all sorts of real world testing scenarios rendering it well suited to be used as part of numerous applications. The proposed model is lightweight and enables real time detection, supporting real time deepfake media detection requiring timely and accurate detection. This capability makes social media monitoring, digital forensics, journalism and public safety practical applications of this capability. The user-friendly interface also improves usability by providing non-technical users an intuitive way to get detection results and visual explanations of detected anomalies helping build trust and transparency for business users. Overall, the proposed hybrid MobileNet-LSTM model makes the contributions of removing the spatial and temporal limitations of deepfake detection systems by using spatial analysis and temporal analysis. And owing to its real time capability, high performance and user friendly design, it is an appropriate tool to combat the persistent problem of deepfake media. Later works may be aimed at incorporating multimodal inputs, e.g. audio and textual analysis, to further enhance detection accuracy and flexibility in dynamic digital environments

REFERENCES

- [1] M. Abdullah, M. Ahmad, and D. Han, "Facial expression recognition in videos: an CNN-LSTM based model for video classification," *2020 International Conference on Electronics, Information, and Communication (ICEIC)*, pp. 1–3, Jan. 2020, doi: 10.1109/iceic49074.2020.9051332.
- [2] A. Sánchez-Caballero, D. Fuentes-Jiménez, and C. Losada-Gutiérrez, "Real-time human action recognition using raw depth video-based recurrent neural networks," *Multimedia Tools and Applications*, vol. 82, no. 11, pp. 16213–16235, Oct. 2022, doi: 10.1007/s11042-022-14075-5.
- [3] A. Mihanpour, M. J. Rashti, and S. E. Alavi, "Human action recognition in video using DB-LSTM and ResNet," *6th International Conference on Web Research (ICWR)*, pp. 133–138, Apr. 2020, doi: 10.1109/icwr49608.2020.9122304.
- [4] W. Ullah, A. Ullah, I. U. Haq, K. Muhammad, M. Sajjad, and S. W. Baik, "CNN features with bi-directional LSTM for real-time anomaly detection in surveillance networks," *Multimedia Tools and Applications*, vol. 80, no. 11, pp. 16979–16995, Aug. 2020, doi: 10.1007/s11042-020-09406-3.
- [5] P. Ghadekar, T. Adsare, N. Agrawal, T. Dharmik, A. Patil, and S. Zod, "Ensemble Learning Approach for Anomaly Detection in Crowd Scene Classification," *IEEE Fifth International Conference on Advances in Electronics, Computers and Communications (ICAEECC)*, pp. 01–11, Sep. 2023, doi: 10.1109/icaeecc59324.2023.10560186.
- [6] M. F. Hashmi, B. K. K. Ashish, A. G. Keskar, N. D. Bokde, J. H. Yoon, and Z. W. Geem, "An exploratory analysis on visual counterfeits using COV-LSTM hybrid architecture," *IEEE Access*, vol. 8, pp. 101293–101308, Jan. 2020, doi: 10.1109/access.2020.2998330.
- [7] X. Yan, X. Gan, R. Wang, and T. Qin, "Self-attention eidetic 3D-LSTM: Video prediction models for traffic flow forecasting," *Neurocomputing*, vol. 509, pp. 167–176, Aug. 2022, doi: 10.1016/j.neucom.2022.08.060.
- [8] M. Jagadeesh and B. Baranidharan, "Facial expression recognition of online learners from real-time videos using a novel deep learning model," *Multimedia Systems*, vol. 28, no. 6, pp. 2285–2305, Jun. 2022, doi: 10.1007/s00530-022-00957-z.
- [9] N. Khairdoost, M. Shirpour, M. A. Bauer, and S. S. Beauchemin, "Real-Time Driver Maneuver Prediction using LSTM," *IEEE Transactions on Intelligent Vehicles*, vol. 5, no. 4, pp. 714–724, Jun. 2020, doi: 10.1109/tiv.2020.3003889.
- [10] M. D. Nguyen, H. N. Vu, D. C. Pham, B. Choi, and S. Ro, "Multistage Real-Time fire detection using convolutional neural networks and Long Short-Term memory networks," *IEEE Access*, vol. 9, pp. 146667–146679, Jan. 2021, doi: 10.1109/access.2021.3122346.
- [11] F. Rezaei and M. Yazdi, "Real-time crowd behavior recognition in surveillance videos based on deep learning methods," *Journal of Real-Time Image Processing*, vol. 18, no. 5, pp. 1669–1679, May 2021, doi: 10.1007/s11554-021-01116-9.
- [12] B. Natarajan et al., "Development of an End-to-End Deep Learning Framework for sign language recognition, translation, and video generation," *IEEE Access*, vol. 10, pp. 104358–104374, Jan. 2022, doi: 10.1109/access.2022.3210543.
- [13] J. Yin et al., "MC-LSTM: Real-Time 3D Human Action Detection System for intelligent healthcare applications," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 15, no. 2, pp. 259–269, Mar. 2021, doi: 10.1109/tbcas.2021.3064841.
- [14] H. E. Dinaki, S. Shirmohammadi, E. Janulewicz, and D. Cote, "Forecasting video QOE with deep learning from multivariate Time-Series," *IEEE Open Journal of Signal Processing*, vol. 2, pp. 512–521, Jan. 2021, doi: 10.1109/ojsp.2021.3099065.
- [15] D.-H. Kang and D.-H. Kim, "1D Convolutional Autoencoder-Based PPG and GSR signals for Real-Time emotion Classification," *IEEE Access*, vol. 10, pp. 91332–91345, Jan. 2022, doi: 10.1109/access.2022.3201342.
- [16] K. Zhang, Y. Li, J. Wang, E. Cambria, and X. Li, "Real-Time video emotion recognition based on reinforcement learning and domain knowledge," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 32, no. 3, pp. 1034–1047, Apr. 2021, doi: 10.1109/tcsvt.2021.3072412.
- [17] H. Nizam, S. Zafar, Z. Lv, F. Wang, and X. Hu, "Real-Time Deep Anomaly Detection Framework for multivariate Time-Series data in

industrial IoT,” *IEEE Sensors Journal*, vol. 22, no. 23, pp. 22836–22849, Oct. 2022, doi: 10.1109/jsen.2022.3211874.

Estimation,” *IEEE Sensors Journal*, vol. 22, no. 14, pp. 13776–13784, Jun. 2022, doi: 10.1109/jsen.2022.3183502.

[18] M. M. E. Yurtsever and S. Eken, “BabyPose: Real-Time Decoding of Baby’s Non-Verbal Communication using 2D Video-Based pose

[19] A. R. Pal, “Classification of pest-infested citrus leaf images using MobileNet V2+ STM based hybrid model,” Doctoral dissertation, Nat.College Ireland, Dublin, Republic of Ireland, 2022